

Annotation-Free Fitness Grading of Nigerian Naira Banknotes via Pretrained Feature Clustering

Abstract—The assessment of banknote fitness, defined as the determination of whether a note is suitable for continued circulation, remains an understudied problem for the Nigerian Naira, where no publicly available fitness-labeled dataset exists and prior work has concentrated on denomination recognition and counterfeit detection. This study proposes an unsupervised approach to Naira fitness grading that operates without fitness annotations. A working dataset of 5,446 images was consolidated from six sources and filtered to 5,349 for surface-based condition analysis. Grayscale images were encoded using a frozen EfficientNet-B0 network pretrained on ImageNet, and the resulting embeddings were grouped into two clusters with k-means, corresponding to the fit and unfit classes defined by the Central Bank of Nigeria's Clean Note Policy. The groupings were evaluated against a balanced set of 200 manually labeled images and achieved a balanced accuracy of 0.950 that was identical across ten clustering runs and under two initialization strategies. A source-controlled evaluation confirmed that the agreement reflected physical condition rather than imaging modality. A comparison with a ResNet-50 backbone under the same pipeline produced substantially lower and less stable agreement, indicating that the separation depends on the choice of pretrained representation.

Keywords—banknote fitness grading, unsupervised learning, Nigerian Naira, convolutional neural networks, transfer learning, k-means clustering, currency condition assessment

I. INTRODUCTION

The capacity of automatically recognizing money is a feature commonly present in today's relevant technologies. These include those used at ATMs (Automated Teller Machine), vending systems, retail self-checkout, counterfeit detection, and for the visually impaired [1], [2]. In Nigeria, cash transactions still dominate despite the introduction of the cashless economy as a result of the 2022 Naira redesign and the Central Bank of Nigeria's 2022 designed policies [3]. Therefore, the need for accurate recognition of the Nigerian Naira is essential. The Naira is available in different polymer and paper denominations, and the notes in circulation are frequently soiled, folded, and distorted, among other defects [4]. Due to similarities in the color and design of several denominations, the Naira presents a difficult recognition problem for both human and automated systems. In order to arrive at a reliable solution, the models developed must be trained on representative data. The need is further reinforced by the more than 2.2 billion people globally affected by visual impairment [5], many of whom rely on assistive systems to perform monetary transactions independently [6].

Prior research on banknote analysis has explored a wide range of approaches, from classical feature extraction methods such as SIFT and SURF [7], to deep learning architectures including convolutional neural networks and lightweight mobile-oriented variants [8], [9]. Several works have addressed the Nigerian Naira specifically. Richard-Nnabu et al. proposed a convolutional neural network for folded Naira denomination recognition to handle different orientations and folds [10], while Tijjani et al. evaluated multiple deep learning architectures on a dataset of 1,808 Naira images and reported the highest performance with MobileNetV2 [11]. Transfer learning approaches built on MobileNetV2 have also been

applied to develop assistive Naira recognition tools for visually impaired users [6], and Faster R-CNN has been used to detect counterfeit higher denominations [12]. A separate but related line of research has examined banknote fitness, namely the assessment of physical condition for the purpose of deciding whether a note should be recirculated or withdrawn from circulation. Pham et al. proposed convolutional neural network methods for fitness classification of USD, INR, and KRW notes using visible-light and near-infrared imagery [13], [14], and earlier work established fitness sorting for the Euro based on soiling features [15]. Nonetheless, these approaches are mainly supervised and rely on expert-annotated fitness labels obtained via specialized imaging hardware. Obtaining this annotation is expensive, currency-specific, and not easily transferable across countries. There is no fitness-labeled dataset that is publicly available for the Nigerian Naira. Previous research on the Naira has focused on denomination recognition and counterfeits. Despite this, the Clean Note Policy of the Central Bank of Nigeria's Clean Note Policy states that the removal of unfit notes from circulation is a national priority [4].

Due to the lack of labeled data as well as the cost of acquiring it, this study investigates methods that do not rely on fitness labels. This study seeks to remedy the gap by using an unsupervised method to grade the fitness of Naira which has been motivated due to the lack of fitness labeled Naira data and the practical issues of producing such labels at scale. The contributions of the work are as follows: (1) an unsupervised fitness grading framework that learns representations of Naira notes from unlabeled images and clusters notes according to their physical condition without expert annotation; (2) the framework was evaluated for stability across ten clustering runs and two initialization strategies, and for sensitivity to the choice of pretrained representation, through a comparison of EfficientNet-B0 with ResNet-50; (3) to perform these experiments, a working image dataset was created through collation of publicly available Naira sources, along with additional images taken by the authors under varied lighting, backgrounds, viewing angles, occlusion, and degrees of note wear. The dataset is unannotated and is intended primarily as the empirical basis for the unsupervised study reported here. Together, these contributions provide an initial step towards annotation-free fitness assessment tools for the Naira and offer a starting point for further research on currency condition assessment in low-resource labeling settings.

II. METHODOLOGY

A. Dataset Preparation

The study of unsupervised fitness grading required a suitably large and varied image dataset that captures the variation encountered in real-world circulation conditions. For this purpose, five public datasets were gathered from available repositories and complemented by a sixth set of pictures collected by the authors, amounting to six sources in total. The constituent sources and their respective image counts are presented in Table I.

TABLE I. COMPOSITION OF THE PREPARED NAIRA IMAGE DATASET

Ref	Source	Repository	Image Count
[16]	Ehimare's Workspace	Roboflow Universe	119
[17]	IMAGEClassification	Roboflow Universe	1,401
[18]	PAU DAT611 Workspace	Roboflow Universe	1,154
[19]	Naira Nigerian Currency Dataset	Kaggle	2,042
[20]	ADEKCURR-DB	Mendeley Data	580
-	Author-Collected Images	Primary Source	150
Total			5,446

The decision to merge multiple sources rather than rely on a single dataset was driven by three considerations. First, each source was acquired under different imaging conditions, which introduces variability in resolution, camera type, and capture distance. Second, the images exhibit uncontrolled backgrounds, inconsistent lighting, and differing orientations, all of which are characteristic of practical deployment environments. Third, stylistic differences across the sources, including variations in note placement, partial occlusion, and the presence of folded or worn currency, further enrich the visual distribution available to the feature extraction and clustering stages. The consolidated dataset has been publicly archived on Kaggle [21], where additional details such as file organization and image specifications are documented. Representative images for each source are shown in Fig. 1.



Fig. 1. Representatives Naira notes from each dataset source.

B. Dataset Filtering and Curation

Prior to feature extraction, the consolidated dataset was subjected to a manual filtering step to remove images deemed

unsuitable for the proposed task. Specifically, images in which human subjects, hands, or feet were prominently visible were excluded from the working dataset. Such images introduce background elements that are incidental to the currency itself and would risk directing the model's attention towards human anatomy rather than the surface characteristics of the banknote. Representatives of the discussed images are shown in Fig. 2.



Fig. 2. Samples of the removed images for the working dataset.

Since the task of grading the fitness relies on detecting damage indicators, such as folds, tears, and soiling on the note surface, prominent human elements constitute a kind of visual noise that can hamper feature extraction. Retaining such images may create misleading associations when extracting features, especially in a denomination-agnostic system. After this filtering step, the working dataset was reduced from 5,446 to 5,349 images, which was used during all following processing stages.

C. Image Grouping and Class Assignment

The image grouping and class assignment was done post filtering. A labeled subset was prepared which served as the evaluation ground truth. Following the recirculation decision of the Central Bank of Nigeria as articulated in its Clean Note Policy, a two-class scheme was adopted. Under this approach, each note is assigned to one of two classes: fit, meaning notes which are suitable for continued circulation, and unfit, meaning notes whose physical degradation is such that they should be withdrawn. The labels were used only for evaluation and were not used in any of the representation learning and clustering.

Notes assigned to the fit class present minimal signs of wear, with no visible tears, perforations, or significant soiling on the note surface. Notes assigned to the unfit class display severe physical degradation, such as deep creases, tears, excessive staining, or structural damage that renders them unsuitable for continued circulation under CBN standards. Notes of intermediate condition, exhibiting early-stage deterioration that does not clearly correspond to either extreme, were retained within the unlabeled working dataset but were excluded from the labeled subset. Consequently, the evaluation measures the separation of clearly fit and clearly unfit notes rather than the full continuum of circulating conditions. Representative images of each class are shown in Fig. 3.



Fig. 3. Image representatives of each class.

The subset of images that was labeled contained 100 images from each class, for a total of 200 images. These images were drawn from the filtered data using stratified sampling across the source repositories. The classes of the data were purposefully balanced so that clustering agreement would be interpretable against a chance baseline of 0.5.

D. Unsupervised Fitness Grading Framework

The fitness grading framework was designed to operate entirely on unlabeled images, such that the class assignments described in Section II-C exerted no influence on the formation of the groupings. An overview of the pipeline is illustrated in Fig. 4. The pipeline consists of three stages: preprocessing, feature extraction, and clustering.

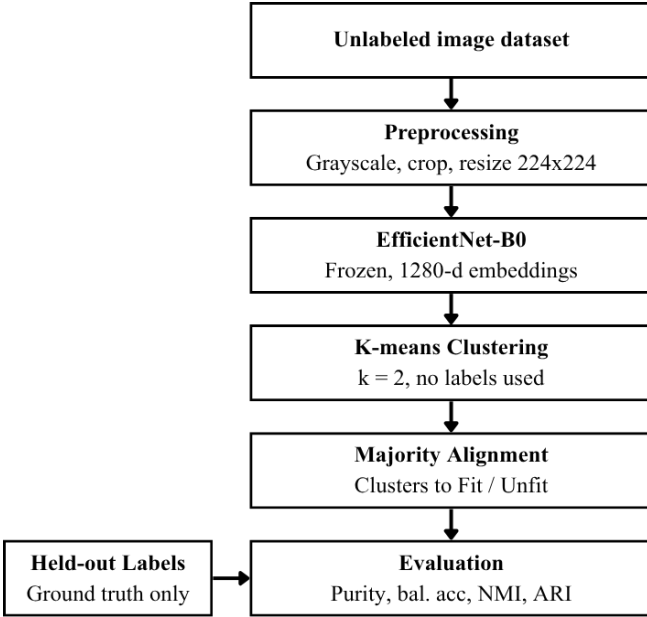


Fig. 4. Overview of the unsupervised fitness grading pipeline.

In the preprocessing stage, each image was cropped to its largest centered square and converted to grayscale, with the resulting single channel replicated across three channels to match the input format expected by the pretrained network. The scanned images sourced from the ADEKCURRE-DB repository [20] were exempted from the cropping step, as the note region was already isolated during digitization. The removal of color information was a deliberate design choice, as Naira denominations are strongly differentiated by hue, and retaining color would risk producing groupings organized by denomination rather than by physical condition. Each image was subsequently resized to a fixed resolution of 224 by 224 pixels.

In the feature extraction stage, EfficientNet-B0 [22], pretrained on the ImageNet dataset [23], was employed as a fixed feature extractor. Each preprocessed image was passed through the network with its classification head removed, and global average pooling was applied to the final convolutional feature maps, yielding a 1280-dimensional embedding of the note's visual appearance. No fine-tuning was performed, and no fitness labels were involved at any point in this stage. The use of transferred representations is motivated by their established ability to capture general texture and structural cues without task-specific supervision [24]. To assess whether the resulting separation depended on the specific pretrained network, a second frozen backbone, ResNet-50 [25], pretrained on the same ImageNet dataset, was evaluated under

an identical pipeline. Its classification head was likewise removed and global average pooling applied, yielding a 2048-dimensional embedding per image in place of the 1280-dimensional EfficientNet-B0 embedding. No fine-tuning was performed for either network.

In the clustering stage, the embeddings were L2-normalized and grouped using the k-means algorithm [26], with the number of clusters set to two in accordance with the fit and unfit scheme. Because k-means is sensitive to its initialization, the clustering was repeated ten times for each backbone, under two initialization strategies, namely k-means++ and random initialization. A single initialization was used per run so that run-to-run variability could be observed directly rather than masked by selection of the lowest-inertia solution. As the feature extraction is deterministic, this initialization is the only stochastic component of the pipeline. Since the cluster indices produced by k-means carry no inherent semantic meaning, each cluster was mapped to a fitness class through majority alignment against the held-out evaluation labels. This mapping serves only to interpret and score the clusters after grouping; the labels exert no influence on how the clusters are formed.

E. Evaluation Protocol

The quality of the discovered groupings was assessed by comparing the cluster assignments against the fit and unfit labels described in Section II-C. Following the majority alignment of clusters to classes, agreement was quantified using five measures: cluster purity, balanced accuracy, the macro-averaged F1 score, normalized mutual information (NMI), and the adjusted Rand index (ARI). Purity and balanced accuracy report the proportion of correctly grouped images, with balanced accuracy averaging the per-class rates to account for class imbalance, while NMI and ARI measure the statistical dependence and the chance-corrected pairwise agreement between the two partitions, respectively. Given the balanced evaluation subset, a purity of 0.5 corresponds to chance-level agreement.

To assess the reliability of the labels, the annotation was performed by the authors according to the criteria of the Clean Note Policy, with disagreements resolved through joint review. In addition, the composition of the evaluation subset across the constituent source repositories was examined to identify any association between class membership and imaging source. Because the scanned ADEKCURRE-DB images were found to be concentrated in the fit class, a source-controlled evaluation was additionally performed in which these images were excluded and the remaining photographed notes were re-scored, separating the contribution of physical condition from that of imaging modality. All five measures were reported as the mean and standard deviation over the ten runs, for each backbone and initialization strategy. The qualitative figures correspond to the EfficientNet-B0 grouping, which was identical across all runs. The outcomes of both the full and source-controlled evaluations are reported in Section III.

III. RESULTS AND DISCUSSION

The pipeline was implemented in TensorFlow and executed on the Google Colab platform using a single GPU. The clustering agreement with the fit and unfit labels is summarized in Table II, reporting both the full evaluation subset and a source-controlled subset from which the scanned images were excluded.

TABLE II. CLUSTERING AGREEMENT OF THE EFFICIENTNET-B0 FIT AND UNFIT GROUPINGS

Set	Images	Purity	Balanced accuracy	Macro-F1	NMI	ARI
Full	200	0.950	0.950	0.950	0.761	0.809
Source-controlled	166	0.952	0.959	0.951	0.765	0.815

^a. Values were identical across ten clustering runs and under both k-means++ and random initialization (standard deviation 0.000)

On the full evaluation subset of 200 balanced images, the EfficientNet-B0 embeddings grouped under k-means produced a purity and balanced accuracy of 0.950, with an adjusted Rand index of 0.809. This result was identical across all ten runs and under both initialization strategies, with a standard deviation of zero, indicating that the grouping is fully determined by the embeddings and is not sensitive to initialization. As the balanced subset corresponds to a chance baseline of 0.5, this represents a substantial agreement between the unsupervised groupings and the labels assigned under the Clean Note Policy, achieved without any fitness supervision.

The confusion matrix is shown in Fig. 5. All 100 fit notes were assigned to the fit cluster, while 90 of the 100 unfit notes were correctly grouped and the remaining 10 were assigned to the fit cluster. The grouping therefore produced no case in which a fit note was placed in the unfit cluster, and the only errors consisted of degraded notes being grouped with fit notes. In the context of the recirculation decision, this asymmetry is relevant, as the method did not erroneously direct sound notes towards withdrawal, although it did retain a small proportion of unfit notes within the fit grouping.

Ground-truth label	Aligned cluster (predicted)	
	Fit	Unfit
Fit	100	0
Unfit	10	90

Fig. 5. EfficientNet-B0's confusion matrix.

Fig. 6 presents two-dimensional t-SNE projections of the evaluation embeddings, colored by ground-truth label and by aligned cluster, respectively. The fit and unfit notes occupy two largely separated regions of the embedding space, and the cluster assignments correspond closely to the ground-truth labels. The few misgrouped notes are located near the

boundary between the two regions, consistent with the gradual visual transition between sound and degraded notes.

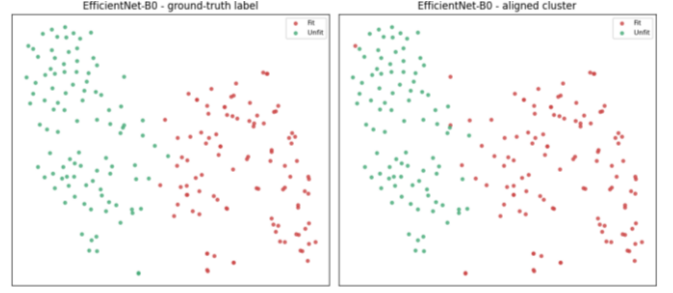


Fig. 6. Two-dimensional t-SNE projection of the EfficientNet-B0 embeddings for the 200-image evaluation subset.

To determine whether the separation depended on the specific pretrained network, the identical pipeline was applied with a ResNet-50 backbone. As reported in Table III, the ResNet-50 embeddings produced markedly lower agreement, with a mean balanced accuracy of 0.703 under k-means++ initialization on the full subset and a best single run of 0.770, and the agreement varied considerably across runs, with a standard deviation of 0.109. Random initialization produced lower agreement still, and several runs were close to the chance baseline of 0.5. None of the ResNet-50 runs approached the agreement obtained with EfficientNet-B0. This contrast indicates that the separation of fit and unfit notes is not a generic property of representations transferred from networks pretrained on natural images, but depends on the choice of representation, and it provides empirical support for the selection of EfficientNet-B0 as the backbone of the proposed pipeline.

To determine whether the lower agreement of the ResNet-50 backbone reflected an absence of cluster structure or a misalignment between that structure and physical condition, the silhouette coefficient was computed for both backbones over the evaluation subset. The ResNet-50 embeddings yielded a silhouette coefficient of $[\approx 0.10]$, which is marginally higher than the $[\approx 0.09]$ obtained with EfficientNet-B0, even though the ResNet-50 balanced accuracy was substantially lower. The two backbones therefore produced clusters of comparable internal separation, yet only the EfficientNet-B0 partition corresponded to the fit and unfit labels. Both coefficients are low in absolute terms, so the relevant observation is the relative ordering, in which the more separated grouping is not the better aligned one. This contrast indicates that the silhouette coefficient reflects the geometric compactness of the clusters in the embedding space rather than their agreement with physical condition. An internal separation measure is consequently not sufficient on its own to confirm that a clustering captures fitness, and validation against the held-out labels remains necessary.

TABLE III. BACKBONE COMPARISON: BALANCED ACCURACY (MEAN $\pm$ SD OVER TEN RUNS)

Backbone (initialization)	Full	Source-controlled
EfficientNet-B0	0.950 $\pm$ 0.000	0.959 $\pm$ 0.000
ResNet-50 (k-means++)	0.703 $\pm$ 0.109	0.668 $\pm$ 0.151
ResNet-50 (random)	0.650 $\pm$ 0.120	0.587 $\pm$ 0.162

^b. EfficientNet-B0 results were identical under both initialization strategies.

When applied to the full working dataset of 5,349 images, the pipeline partitioned the notes into two groups, the sizes of which are shown in Fig. 7.

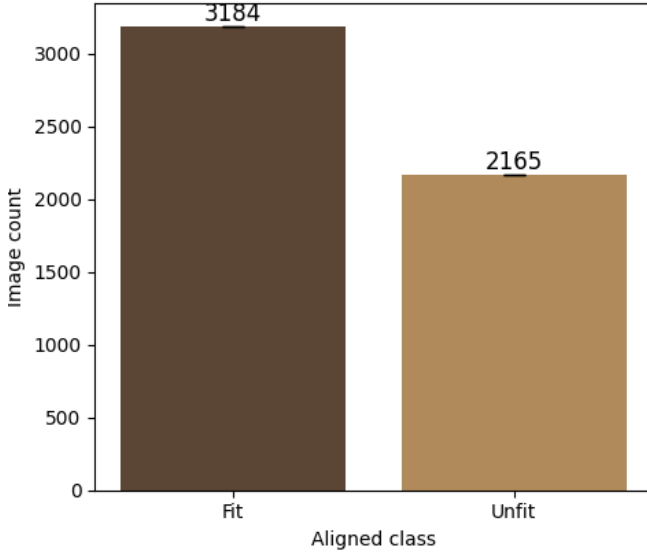


Fig. 7. Distribution of images across the aligned fit and unfit clusters for the full filtered dataset of 5,349 images.

Representative images drawn from each group are presented in Fig. 8, where the visual distinction between minimally worn and degraded notes is apparent. Because ground-truth labels were available only for the 200-image evaluation subset, the grouping of the remaining images constitutes an extrapolation that was not directly validated. The silhouette coefficient computed over the full dataset was 0.091, which indicates that the two groups are not sharply separated in the embedding space when the entire range of circulating conditions is considered. This is consistent with fitness varying as a continuum rather than as two discrete categories, and it aligns with the decision to evaluate on clearly fit and clearly unfit notes.

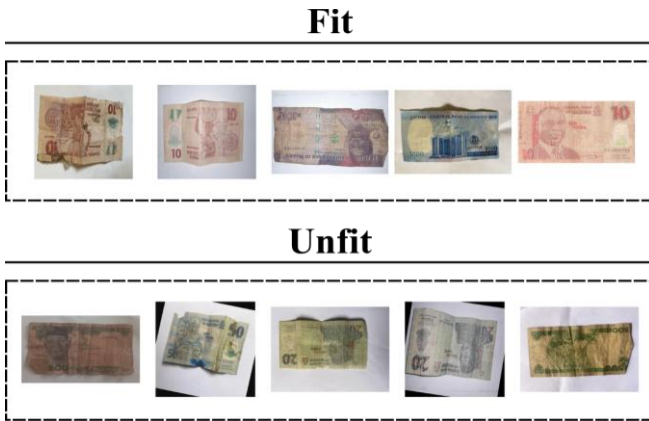


Fig. 8. Representative images drawn from the aligned fit and unfit clusters produced by the unsupervised pipeline.

The findings show that a transfer of representations from a network trained on natural images, combined with a simple clustering procedure, suffices to separate clearly fit from clearly unfit Naira notes without any fitness annotation. The grayscale helped direct the result towards surface condition representation. The result's sensitivity to the backbone in Table III indicates that the quality of the separation is governed by the choice of representation rather than by the clustering procedure. The comparable silhouette coefficients

of the two backbones, reported in Section III, further indicate that geometric separation in the embedding space does not by itself imply separation by physical condition, so an internal cluster-quality measure cannot substitute for evaluation against fitness labels.

Several limitations qualify these findings. First, the evaluation excluded notes of intermediate condition, so the reported agreement reflects the separation of clearly fit and clearly unfit notes rather than performance across the full range of circulating conditions, and agreement on borderline notes is expected to be lower. Second, the labeled subset was small, comprising 100 images per class, which yields wide confidence intervals around the reported measures. Third, although the source-controlled evaluation indicates that imaging modality did not drive the result, the labeled subset was not balanced across source repositories, and a larger source-balanced evaluation set would provide a stronger basis for assessment. In addition, the agreement was obtained with a specific pretrained backbone, and the comparison with ResNet-50 shows that the quality of the separation depends on the choice of representation. The approach therefore should not be assumed to transfer to arbitrary pretrained networks without re-evaluation. Finally, the grouping of the full dataset beyond the labeled subset was not validated, and the labels were assigned by the authors according to policy criteria rather than by certified currency examiners.

IV. CONCLUSION AND RECOMMENDATIONS

This study examined an unsupervised approach to fitness grading of Nigerian Naira banknotes, motivated by the absence of any publicly available fitness-labeled Naira dataset. Grayscale images from a consolidated set of 5,349 images were encoded using a frozen EfficientNet-B0 network and grouped into fit and unfit clusters with k-means. The groupings achieved a balanced accuracy of 0.950 against a balanced labeled set, and a source-controlled evaluation confirmed that this agreement reflected physical condition rather than imaging modality, indicating that transferred representations with clustering can separate clearly fit from clearly unfit notes without fitness supervision. The grouping was stable across repeated runs and initialization strategies, whereas a comparison with a ResNet-50 backbone showed that this separation depends on the choice of pretrained representation.

Future work should expand and source-balance the labeled set with expert annotation, extend the evaluation to intermediate-condition notes across the full continuum of circulating conditions, and investigate domain-specific representations such as self-supervised pretraining or surface-damage features to improve separation beyond that obtained from networks pretrained on natural images.

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